

BrickAtlas: Measuring Visual, Structural, and Executable Reasoning on Shared Brick Assemblies

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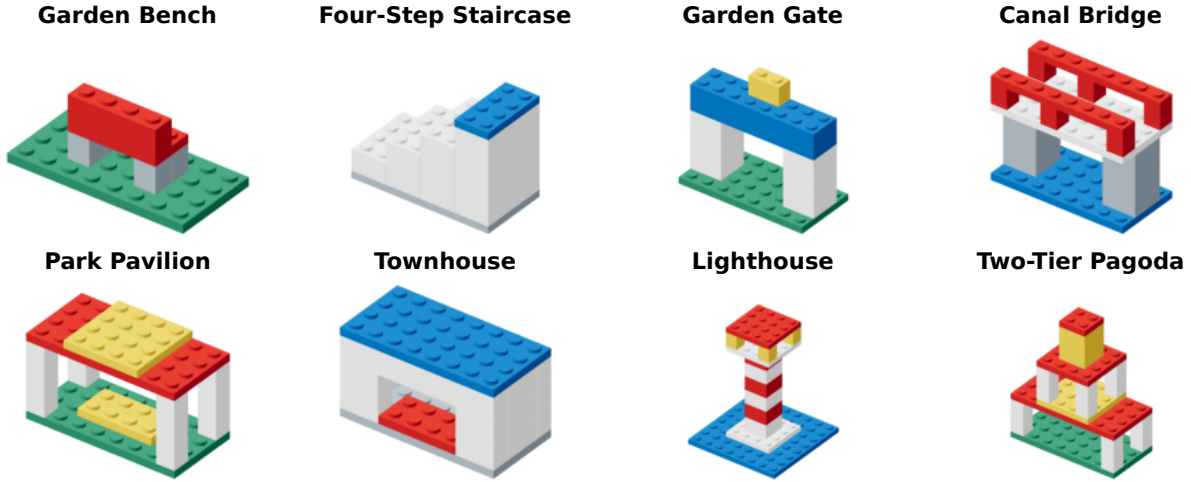


Figure 1. **Curated BrickAtlas calibration cases.** Eight of twelve original, explicitly authored structures span furniture, architecture, infrastructure, and landmarks at easy, medium, and hard difficulty. Every model uses at most four role-based colors and has verified geometry and an executable assembly order. This curated layer is separate from the 5,120-source procedural population.

Abstract

Understanding a brick assembly requires distinguishing its appearance, hidden structure, requested changes, and executable construction. BrickAtlas connects eight task families through 5,120 procedural source assemblies and 117,910 task-condition cases with explicit input privileges and typed success criteria. A separate twelve-object curated layer supplies coherent task exemplars at three difficulty levels. We add a finite-grammar ambiguity track: 600 paired conditions have exhaustively checked answer sets, and 90 exterior-only queries are genuinely undetermined within the declared grammar. A separate renderer audit verifies identical exterior pixels across admissible layouts. An eight-model development matrix and same-source algorithmic controls expose different failure patterns, while full-casebank copying achieves 87.1% completion F1 and 97.4% removal F1 without completing either task. Ground-truth and execution visualizations make these distinctions inspectable. We release replayable outputs, source-level audits, and an explicit experimental matrix. The model sample

is small; independent confirmation, richer licensed objects, and real human metric calibration remain outstanding.

1. Introduction

A system can recognize the bricks in an object without recovering their poses, match its exterior without identifying hidden connections, or return a nearly unchanged structure with high similarity after failing an edit. A correct final assembly also need not make a submitted construction order executable. Treating these outcomes as a single notion of “structural intelligence” obscures what a model accomplished.

BrickAtlas uses shared source assemblies to separate these endpoints. Figure 3 shows actual inputs and ground truth, while Fig. 5 distinguishes a valid sequence from a dead end. The benchmark records what information is provided, what output is required, and which equivalences the evaluator accepts. A correct answer may be a structure, a relation tuple, an executable permutation, or a set-valued statement that the available observation is insufficient.

This is not a claim to originate brick generation, reconstruction, or multi-model assembly evaluation. BrickGPT [10] and BrickNet [6] provide strong generative foundations; MEPNet [14], Neural Assembler [15], and TreeSBA [3] address visual assembly. BC-Bench [7] directly evaluates multimodal models on brick selection and single-step pose estimation. Our narrower opportunity is *measurement validity across shared objects*: when does a plausible appearance, a high similarity score, or one valid reference justify a downstream structural conclusion?

We investigate three questions. **RQ1**: Does global structure similarity indicate that a requested change was actually performed? **RQ2**: Can an observer determine a hidden relation from the provided evidence, and how should an evaluator score genuine ambiguity? **RQ3**: Is candidate selection stable to reordering beyond instability under identical repetition? Negative answers are informative benchmark findings, not failures to produce a preferred model ranking.

The release contributes (1) a source-grouped eight-family evaluation contract with visual, structural, and executable endpoints; (2) a new finite-grammar, observation-conditioned relation track with exhaustive ground-truth sets and public-input enumeration; and (3) full-population short-cut audits, an eight-model descriptive matrix, and paired protocol controls. We separate implemented capability, measured development evidence, and unrun experiments throughout.

2. Related Work and Position

Generation and assembly. BrickGPT [10] generates stable and buildable objects from text. BrECS [1] studies budget-aware generation and completion with efficient constraint checking. BrickNet [6] uses human-designed LDraw objects and typed connectors; its 320,808 samples and 9,743 part variants far exceed our grid vocabulary. Nominal contact in BrickAtlas is not a substitute for their structural or physical scope. MEPNet [14] reads stepwise manuals; Neural Assembler [15] predicts components, topology, poses, and sequence from multiple views; TreeSBA [3] uses a tree representation and synthetic-to-real supervision. Break and Make [13] studies interactive inspection, disassembly, and rebuilding. Their input protocols differ from fixed image or symbolic submissions and require audited adapters for comparison.

The closest multimodal benchmark. BC-Bench/BrickComposer [7] evaluates diverse-brick selection and pose estimation from manual context and current state, including feedback and training studies. Thus “many models assembling bricks” is not sufficient novelty. BrickAtlas instead makes acceptable alternatives, changed-region preservation, current-ID localization, and observation-conditioned answer sets explicit across tasks. Table 1 states the distinc-

tion without claiming superior diversity or equating different sample units.

Diagnostic visual reasoning. CLEVR [5] establishes controlled compositional reasoning evaluation; BLINK [2] tests perceptual skills, including multi-view and spatial reasoning. CSGNet [12] predicts executable shape programs. SpatialBabel [8] examines reconstruction across scene-code languages. Output-protocol effects are therefore not new in themselves. Similarly, MCQ position bias is established [16]; our candidate experiment specifically compares actual pose changes against an identical-repeat baseline, without inferring internal model mechanisms from external API replies.

3. Benchmark Construction and Tasks

3.1. Geometry, Sources, and Splits

A part is (i, t, c, x, y, z, r) : instance ID, type, color, minimum body corner, and quarter-turn yaw. Y is **vertical**; X, Z use 8 mm studs and Y uses 3.2 mm plate heights. Origins are integer coordinates in $[0, 24]^3$, bodies must fit in $[0, 32]^3$, and submissions contain at most 64 parts. Odd yaw swaps width and depth. Geometry is nominally valid if bodies do not overlap and every elevated part has at least one directly supporting stud. Force balance, clutch, arbitrary CAD geometry, and robotic grasping are outside this contract.

Eight seeded policies accept 640 geometrically unique sources each. They vary attachment height, branching, thin parts, columns, alternating beams, explicit paired supports, and enclosed interiors. Every source starts from a 4-by-8 plate, a deliberate control and a diversity limitation. Canonicalization removes color and translation and considers four global yaw rotations; exact duplicates within this release and against two earlier releases are rejected. Mirrors are not equivalent.

Six policies use a deterministic geometry-hash split, while bridge and shell are policy-held-out (Table 2). All task variants and prefixes retain the source group. This is a procedural-policy holdout, not absence of shared primitives or a semantic generalization guarantee. An exhaustive 8,305,836-pair cross-split audit finds no canonical-token Jaccard similarity at least 0.8; it does not test best-aligned subassemblies or pretraining contamination.

The source census measures 9–47 parts per assembly and **19 assembled types from a 25-type catalog**. Six catalog types occur only in recognition. Size bins contain 1,445 small (≤ 16), 2,141 medium (17–28), and 1,534 large (> 28) objects. Policy and size are not independent; full distributions, missing strata, and catalog counts are provided in the supplement. More views or task conditions do not create more independent objects.

Work	Supplied information	Main output / setting	BrickAtlas distinction
BrickGPT; BrECS	Text or target constraints	Generation / completion	Multi-task measurement, not new generation
BrickNet	LDRAW parts and typed connectors	Graph-backed build programs	Smaller domain; explicit answer equivalence
MEPNet; TreeSBA	Manuals or multi-view images	Poses / assembly actions	Shared task labels and changed-region metrics
Break and Make	Interactive observations	Disassembly and rebuilding	Fixed-input and tool tracks kept separate
BC-Bench	Manual, current state, candidate bricks	Selection and single-step pose	Hidden-layout ambiguity and typed multi-task success
SpatialBabel	Images and scene-code languages	Primitive scenes / spatial QA	Nominal contacts, edits, and original-order execution

Table 1. Closest-work positioning by input and output protocol. This is not a capability checklist or an empirical comparison on matched data. Specialized-method adaptation remains unrun.

Split	Sources	Task conditions
train	2,690	61,870
validation	392	9,016
test_id	758	17,584
test_ood	1,280	29,440
Total	5,120	117,910

Table 2. Source-group split. ID cases additionally include 150 isolated recognition questions, not 150 new assembly objects.

Curated calibration layer. The procedural population provides controlled scale but often lacks a recognizable object silhouette. We therefore add twelve explicitly authored, CC0 cases without changing v2 or its historical scores: four easy, four medium, and four hard structures across furniture, architecture, infrastructure, and landmarks (Fig. 1). Each has two to four functional color roles, exact structure JSON, four views, occupied-layer renders, exploded ground truth, and verified assembly checkpoints. Eight task demonstrations bind one such object to each task contract. This is a qualitative/calibration layer, not twelve independent semantic test families or a substitute for licensed human-designed data.

3.2. Task and Information Contract

Table 3 defines the task families. **Recognition** predicts a known catalog type, color, and stud count from isolated views. **Relations** uses a symbolic assembly and two IDs to ask direct contact, vertical ordering, mating-stud count, and shortest undirected contact-graph distance; a non-contact pair can still have a graph path. **Reconstruction** predicts a full structure from reference information and BOM. **Generation** accepts any legal structure satisfying exact extents, zero anchoring, a piece budget, color count, connected component count, and minimum maximum contact degree. It

is constrained layout generation, not semantic text-to-shape evaluation.

Completion removes the last $\lfloor n/4 \rfloor$ pieces of a valid insertion sequence and requires preserving the remaining prefix. It does not cover arbitrary holes. **Editing** rotates the whole structure, replaces all occurrences of one color, or removes one topmost piece. Rotation maps positive X toward positive Z , keeps Y , and reanchors minimum X, Z . **Repair** has no-fault, color-fault, and one-stud-shift cases on exposed topmost pieces. It requires corrected geometry and faulty current IDs, with an empty fault list for a normal input. **Planning** supplies the target with IDs in lexical order and requests any executable permutation for vertical assembly/disassembly.

Each source yields 13 variants and 23 information-conditioned cases. Reconstruction, completion, and all three repairs have three reference conditions. With 25 catalog types in six colors, the total is $5,120 \times 23 + 150 = 117,910$. **Ordinary RGB** provides isometric/top/front/side assembled views and BOM; interiors remain hidden. **Layer disclosure** provides an assembled view and top images of each bottom- Y layer with other layers removed. **Symbolic reference** supplies complete geometry. The latter two add information and require full type/color/pose recovery. Other families use their declared symbolic or constraint inputs. Fault labels, changed IDs, and private render recipes are excluded from ordinary prompts.

4. Ground Truth and Measurement

4.1. Visual Compatibility Versus Exact Structure

Ground-truth records bind input hashes, source groups, target/source geometry, protected and changed IDs, contacts, nominal stud mates, executable witnesses, and metric applicability. Multiset matching ignores arbitrary output IDs and symmetric per-part yaw, but does not silently align global

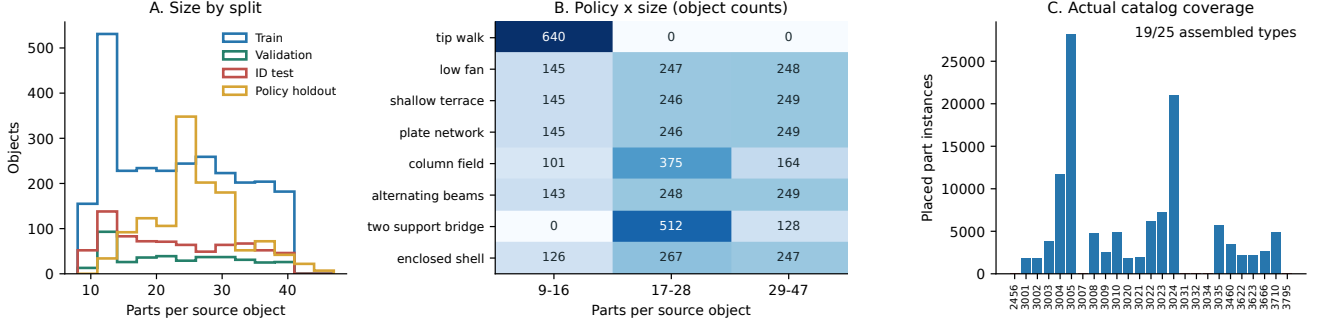


Figure 2. Full source census: size distributions, policy-by-size coverage, and actual part instances. Zero strata and unused catalog types remain visible rather than being hidden by derived task counts.

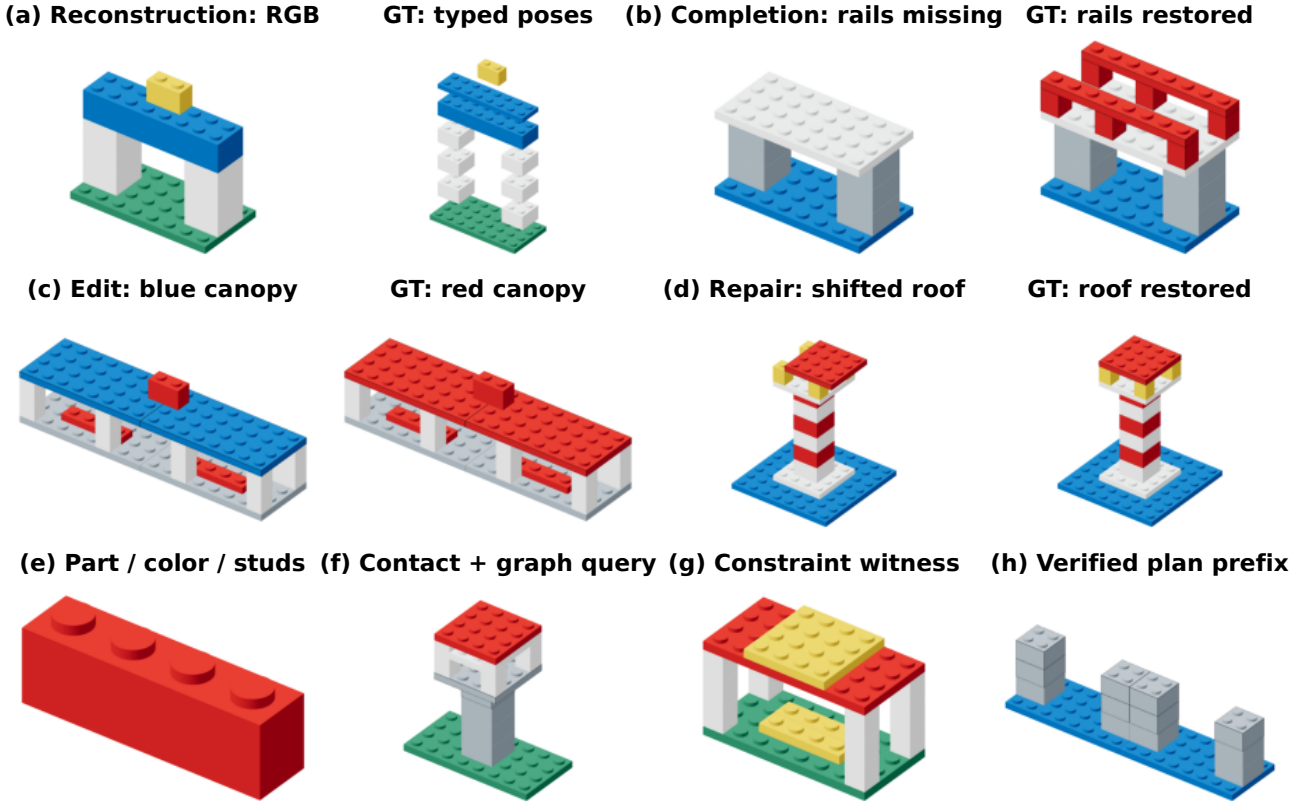


Figure 3. **Task-specific curated input and ground truth.** Reconstruction maps gate views to typed poses; bridge completion restores rails; station editing changes only the canopy color; lighthouse repair restores the shifted roof. Bottom panels illustrate part recognition, a typed relation query, one valid generation witness, and a verified planning prefix. Exploded offsets are display-only, not valid poses or ordinary model observations.

translation or rotation. For m matched parts in target T and answer A ,

$$P = m/|A|, \quad R = m/|T|, \quad F_1 = 2m/(|A| + |T|). \quad (1)$$

BOM F1 removes pose; occupancy IoU removes part boundaries; edge F1 requires matched endpoints and mating-stud counts.

For ordinary reconstruction, let S^+ be colored first-hit cells along positive X, Y, Z . We accept

$$V(A, T) = \mathbf{1}\{\text{valid}(A)\} \mathbf{1}\{\text{BOM}(A) = \text{BOM}(T)\} \cdot \mathbf{1}\{S^+(A) = S^+(T)\}. \quad (2)$$

This proxy allows alternative hidden arrangements; it is not

Family (variants)	Public input	Output	Success contract	Cases
Part recognition	2 isolated RGB views	Type, color, stud count	All three fields correct	150
Relations (2)	Symbolic structure + two IDs	Connection, above, studs, distance	All four fields correct	10,240
Reconstruction	Reference + BOM	Full structure	Condition-specific match + legality	15,360
Generation	Six public constraints	Any satisfying structure	All constraints + legality	5,120
Completion	Current prefix + reference + BOM	Full structure	Match + preserved prefix	15,360
Editing (3)	Current structure + operation	Full edited structure	Exact edit + protected preservation	15,360
Planning (2)	Target structure + direction	Permutation of all IDs	Complete executable order	10,240
Repair (3)	Current + reference + BOM	Structure + faulty current IDs	Match + preservation + localization	46,080

Table 3. Eight task families. “Reference” denotes the declared RGB, layer, or symbolic condition. Symbolic tasks are not visual scores.

arbitrary camera visibility or pixel similarity. Full-reference correctness is reported separately. Completion subtracts preserved source matches before computing *addition* F1. Edits/repairs require protected piece preservation; repairs additionally require faulty-ID localization. False alarms are defined only for no-fault cases.

RQ1: no-operation shortcuts. Copying a prefix with k missing parts from an n -part target gives

$$F_1^{\text{copy}} = \frac{2(n-k)}{2n-k}, \quad F_{1,\text{added}}^{\text{copy}} = 0. \quad (3)$$

Copying before a one-piece removal similarly yields $2(n-1)/(2n-1)$, approaching one without performing the edit. A full 35,840-case no-operation census confirms this: 87.1% completion and 97.4% removal F1, both with zero task success. Positive repair copies reach 94.9% F1 without correct localization. Normal no-fault repair correctly succeeds unchanged. All seven endpoints contain one case per source; per-case results and complete global/local metrics remain available.

4.2. An Ambiguity-Aware Extension

Matching appearance need not identify hidden relations. Instead of forcing one reference answer, we make the admissible set explicit. The new, separately versioned track uses sealed shells with a $2 \times W$ internal layer, $W \in \{2, 4, 6\}$, tiled by identical 1-by-2 bricks. Three stack heights give nine parameterized families. The tilings are exhaustive within this finite grammar: their counts are 2, 5, and 13, independently checked by the recurrence $c_W = c_{W-1} + c_{W-2}$, $c_0 = c_1 = 1$. There are 60 valid layout states across the nine families.

For each pair of blue upper bricks we ask whether they share a directly supporting red brick. Given the observation

Condition	Groups	Cases	Yes	No	Undet.
Exterior	9	300	0	210	90
Support disclosed	9	300	36	264	0

Table 4. New finite-grammar task counts, separate from v2. Nine parameterized families are not nine independent semantic sources.

O and public grammar $\mathcal{G}$, let $\mathcal{A}(O, \mathcal{G})$ be all admissible layouts. With Boolean query q ,

$$y(O) = \begin{cases} \text{yes}, & \forall A \in \mathcal{A}, q(A) = 1, \\ \text{no}, & \forall A \in \mathcal{A}, q(A) = 0, \\ \text{undetermined}, & \text{otherwise.} \end{cases} \quad (4)$$

Exterior and support-disclosed conditions create 600 paired cases (Table 4). Ninety exterior questions are genuinely ambiguous within the declared grammar; disclosure resolves them. Known geometry and the grammar are supplied symbolically, so this extension tests observation-consistent structural inference, not pure visual perception. Public records omit labels and hidden supports; an enumeration solver consumes only these public inputs and solves all 600 cases.

All layouts agree in BOM, colored occupancy, and declared surfaces and pass two geometry implementations. A separate high-resolution renderer checks all 60 layouts in four views: 240 rendered frames match their within-family reference pixel-for-pixel. This result is renderer/camera-specific, not human calibration. Unweighted always-no accuracy is 70% exterior and 88% disclosed; class-balanced accuracy is 50% in both. We therefore report balanced success, false certainty on ambiguous cases, and unnecessary abstention on determinate cases. Complete baseline and planned VLM tables are in the supplement.

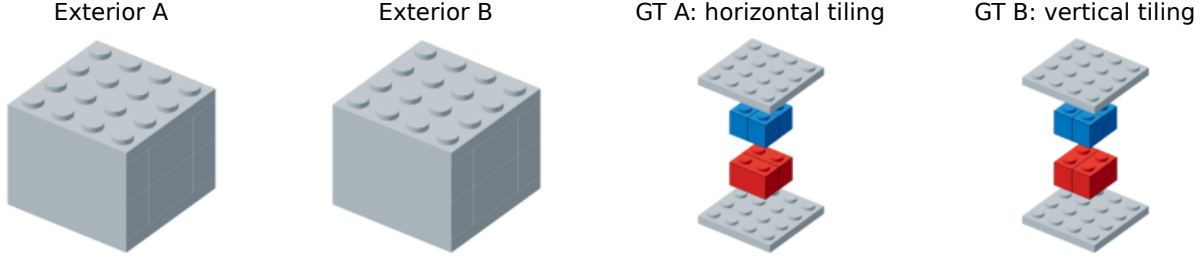


Figure 4. **Same exterior, different ground truth.** The two sealed structures have identical rendered exterior pixels under the tested cameras. Exploding their layers reveals horizontal versus vertical red internal bricks; side walls are omitted in the disclosed views. The outlined adjacent blue pair shares one red supporter in A but not B. The exterior-only answer is therefore *undetermined*; disclosing the support layer makes the answer definite. Exploded views are evaluator illustrations.

4.3. Execution and Evaluation Integrity

Execution stops at the first unknown, duplicate, unsupported, colliding, or blocked action. We record legal-prefix fraction, coverage, duplicate status, and the zero-based failure step. An incomplete legal prefix fails success but has no illegal step. Height sorting solves all 10,240 plans in this vertical domain: supporters are lower and already present; an earlier part overlapping the footprint must end below the new part, or the final geometry would collide. Reversing the sequence gives disassembly. This establishes a constraint-following ceiling, not difficult search.

Frozen selection manifests reject unknown/duplicate predictions; missing/malformed submissions remain in applicable denominators. Null means inapplicable, not a removed failure. A versioned strict decoder closes legacy primitive-type coercion without altering recorded scores. All 117,910 oracle answers pass; independent metrics and mutation checks are detailed in the supplement. Replay labels model, harness, environment, and evaluator operations separately. Visual unfolding of an unordered structure is never called a model plan.

5. Models, Baselines, and Experiments

5.1. Eight-Model Development Matrix

We evaluate the same 39 default/ordinary task cases from three already-exposed development objects (12/20/29 parts) with eight models. Four models retain their original verified responses; four new models receive identical prompts, image bytes, and the 2,200-token JSON output cap. Temperature is zero, model order rotates by case, automatic retries and provider fallback are disabled, and provider, tokens, truncation, latency, and settled charges are retained. Gemini reasoning is disabled; other models have no requested reasoning budget. Cross-round time/provider differences remain potential confounds.

The main matrix groups variants only within each task family. The supplement breaks out all 13 variants to expose

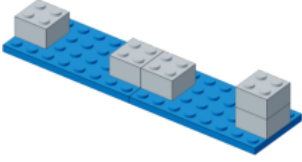
floors and ceilings. Same-source controls include copying current geometry, train-only BOM/extent retrieval, symbolic rules, greedy legal actions, and height sorting. Unsupported method/task pairs are not scored as failed attempts. Table 6 reports local metrics alongside exact success so schema failure, pose error, and doing nothing are not conflated. Global scores alone cannot identify a model’s internal perceptual or reasoning failure. In this shared sample, 3 of 13 variants have zero exact success across all eight models and 0 are saturated. Reconstruction successes range from 0/3 to 0/3; mean added-part F1 ranges from 0.0% to 11.1%. These are descriptive ranges, not evidence of a stable population ranking. Floor/ceiling patterns motivate calibrated difficulty and local metrics before a larger comparison.

5.2. Paired Tools, Representations, and Training

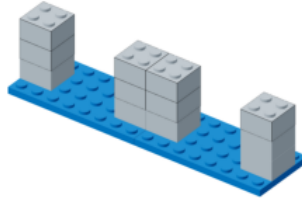
A preceding 24-source, two-model study contains 674 one-shot calls and 56 paired controls. It is not pooled with the three-source matrix. Both models fail exact ordinary reconstruction while succeeding on all recolor/removal cases. Eight-source Gemini controls reuse the same first answer for reflection and target-blind validation (Table 7). Symbolic reference supplies privileged geometry; its gain is not a pure perception intervention. A zero exact-success floor under missing/mismatched images also cannot establish that images are ignored.

The local SmolVLM-256M [9] pilot includes base and three seeds each of single-task, multi-task, and leave-edit LoRA [4]. All ten jobs evaluate the same 337 cases. Supervised answer-token budgets are matched, not total compute. No positive edit transfer is observed, and all base outputs fail JSON parsing or schema checks. Original predictions and earlier Qwen3 [11] pilots remain in the supplement. These negative results motivate an effective baseline before a larger transfer study; adapters existing on disk do not establish learning.

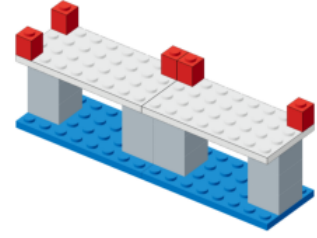
Legal prefix: 7/28 parts



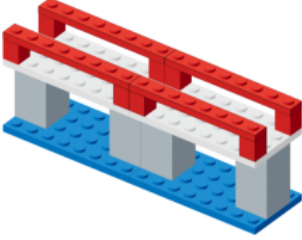
Legal prefix: 14/28 parts



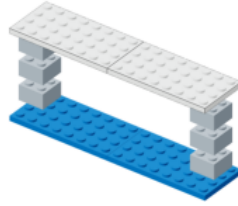
Legal prefix: 21/28 parts



Legal prefix: 28/28 parts



Invalid strategy: decks too early



Blocked inner pier outlined

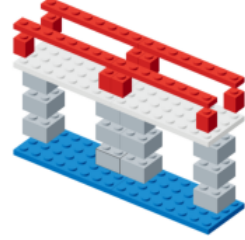


Figure 5. **Ground-truth construction versus a designed dead end.** Four snapshots follow the curated double-span bridge’s verified height-sorted sequence. The last two panels contrast a legal prefix that places both decks before the inner piers with the exploded target; the missing inner pier can no longer descend vertically. The evaluator retains the submitted order and first rejection rather than repairing it.

	Symbolic	RGB	Constraints	RGB	Symbolic	RGB			
Model / algorithm	Relations	Recon.	Gen.	Complete	Edit	Plan	Repair	Format	USD
Vision-language models: same three source objects									
GPT-4.1 mini	3/6	0/3	0/3	0/3	6/9	2/6	4/9	38/39	0.071
GPT-4.1	1/6	0/3	0/3	0/3	6/9	2/6	2/9	35/39	0.341
Gemini 2.5 Flash	2/6	0/3	0/3	0/3	6/9	2/6	3/9	34/39	0.089
Qwen3-VL-32B	0/6	0/3	0/3	0/3	6/9	0/6	1/9	22/39	0.025
Claude Haiku 4.5 ⁺	0/6	0/3	0/3	0/3	6/9	0/6	2/9	22/39	0.317
Gemini 3 Flash (preview) ⁺	3/6	0/3	1/3	0/3	6/9	4/6	1/9	27/39	0.144
Qwen3-VL-8B ⁺	1/6	0/3	0/3	0/3	6/9	0/6	1/9	26/39	0.025
Llama 4 Maverick ⁺	0/6	0/3	0/3	0/3	5/9	1/6	1/9	39/39	0.037
Public-input algorithmic controls: supported tasks only									
Copy current	–	–	–	0/3	0/9	–	3/9	–	–
BOM / extent retrieval	–	0/3	3/3	0/3	–	–	0/9	–	–
Symbolic rules	6/6	–	–	–	9/9	–	–	–	–
Greedy legal action	–	–	–	–	–	5/6	–	–	–
Height-sorted plan	–	–	–	–	–	6/6	–	–	–

Table 5. **Main descriptive matrix: eight models, identical three-source tasks.** Entries are exact successes / cases; variants share sources, so denominators are not independent trials. ⁺ denotes the four newly run models. Algorithm rows use exactly the same cases on supported tasks; – means unsupported or no API charge field, not zero success. No pooled rank or significance claim is supported by three objects. Recognition is separate in the 24-source study because this shared screen has no isolated-part cases.

5.3. Candidate Consistency and a Visual Control

Coordinate-normalized two-piece probes compare full re-construction, local pose, candidate selection, and privileged symbolic answers. Candidate lists add information, so they are not a pure output-length change. A 12-source, four-

model follow-up explicitly distinguishes reordering from identical repetition. For actual selected poses a_g, a_g^r, a_g^π , the paired endpoint is

$$D_g = \mathbf{1}[a_g^\pi \neq a_g] - \mathbf{1}[a_g^r \neq a_g]. \quad (5)$$

Model	Reconstruction		Completion		Planning	Repair	Truncated
	Part F1	Surface F1	Added F1	Preserve	Legal prefix	Fault F1	
GPT-4.1 mini	4.3	18.6	0.0	98.5	49.3	33.3	0/39
GPT-4.1	2.4	18.2	0.0	100.0	51.8	16.7	0/39
Gemini 2.5 Flash	5.8	29.0	11.1	66.7	44.3	27.8	1/39
Qwen3-VL-32B	0.0	0.0	0.0	33.3	1.4	0.0	3/39
Claude Haiku 4.5	1.1	16.1	0.0	100.0	19.7	0.0	2/39
Gemini 3 Flash (preview)	24.7	35.4	11.1	66.7	78.2	16.7	0/39
Qwen3-VL-8B	0.0	0.0	0.0	33.3	0.0	0.0	3/39
Llama 4 Maverick	2.4	15.9	0.0	100.0	39.6	16.7	0/39

Table 6. Task-specific diagnostics on the same three sources, percentages except truncation counts. All returned responses remain in each applicable mean, including format failures. Fault F1 includes only the six faulty repair cases, excluding the three normal controls.

Paired condition	Original	Intervention
no-image	0/8	0/8
mismatched-images	0/8	0/8
symbolic-reference	0/8	8/8
reflection	0/8	0/8
validation	0/8	0/8
repeat	0/8	0/8
relative	6/8	0/8

Table 7. Measured eight-source Gemini controls. Relative denotes assembly representation; other rows are reconstruction. These are a different cohort from Table 5.

Mean excess mismatch is 0.50–0.75 across models, with all 192 requests retained. Source-bootstrap intervals and per-model counts are in the supplement. Mismatch is inconsistency, not incorrectness, and the same 12 sources do not become 48 independent objects.

A known-renderer analysis-by-synthesis control solves all 16 original two-piece sources across two development studies and preserves their selected pose under permutation. It uses public BOM, candidate poses, and rendered mask matching, not hidden target geometry. Repeated conditions yield 44/44 image-conditioned successes; all 16 no-image cases abstain. Exact renderer access, canonical colors, and candidate enumeration favor this method, so it is a signal control, not a general learned reconstruction baseline.

6. Experimental Coverage and Remaining Work

The supplement now contains explicit tables for twelve prospective experiments, with all unrun result cells blank. They cover independent main evaluation, observation ablations, candidate robustness/debiasing, difficulty/camera/palette factors, external sources, human calibration, training/transfer, equal-budget feedback tools, representation-by-decoder factorials, the new ambigu-

ity track, harder planning domains, and task-adapted cross-benchmark methods. MEPNet, Neural Assembler, TreeSBA, BrECS, BrickGPT, BrickNet, and BrickComposer appear as planned adapters, not copied published scores.

The next sequence is to acquire actual licensed external structures, validate mappings and exclusions, calibrate acceptance with real independent reviewers, establish effective specialist baselines, and then freeze new source selections and analysis. Sample size follows precision, not a desired ranking. For incomplete paired experiments, complete-case effects need coverage and worst-case missing-outcome bounds. Existing public development cases cannot be relabeled independent hidden tests.

7. Limitations and Conclusion

BrickAtlas is currently a controlled grid benchmark, not a comprehensive semantic or physically validated assembly benchmark. Its common base, 19 assembled types, suffix-only completion, topmost repairs, and height-solvable planning restrict coverage. The ambiguity extension is exhaustive only within its stated grammar; nine parameter families are not independent external sources. Pixel equality is verified on one renderer, not every camera or human observer. Current human-review packets contain no real submitted judgments. Small adaptive model studies and same-machine replay do not replace independent confirmation and second-environment reproduction.

Within this scope, the benchmark makes useful distinctions executable: changed-region success is not global similarity; an admissible structure does not determine every hidden relation; and a plausible final object does not certify a submitted plan. The release combines real 3D ground-truth illustrations, explicit observation-conditioned answers, algorithmic controls, and measured model outputs. Stronger claims require the remaining external and human evidence, not merely more task variants or a denser table.

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